Comparison with at Least Two Peer-Reviewed Published Accounts

Paper 1

Citation:

Pyron, R. A., & Wiens, J. J. (2011). A large-scale phylogeny of Amphibia including over 2,800 species, and a revised classification of extant frogs, salamanders, and caecilians. *Molecular Phylogenetics and Evolution, 61*(2), 543–583. https://doi.org/10.1016/j.ympev.2011.06.012

Comparison:

The phylogenetic tree generated in this laboratory is generally consistent with the findings of Pyron and Wiens (2011), who reported that closely related frog species tend to cluster according to their evolutionary history inferred from molecular data. In both analyses, species belonging to the same or closely related genera, such as *Pelophylax*, *Lithobates*, and *Hyla/Dryophytes*, form distinct evolutionary groups. Although the present study includes only a limited number of species and uses a single mitochondrial marker (COI), the observed clustering agrees with the broader phylogenetic relationships reported by Pyron and Wiens. Minor differences in branching order may be attributed to the smaller taxon sampling and the use of only one gene sequence in the present analysis.

Paper 2

Citation:

Frost, D. R., Grant, T., Faivovich, J., Bain, R. H., Haas, A., Haddad, C. F. B., de Sá, R. O., Channing, A., Wilkinson, M., Donnellan, S. C., Raxworthy, C. J., Campbell, J. A., Blotto, B. L., Moler, P., Drewes, R. C., Nussbaum, R. A., Lynch, J. D., Green, D. M., & Wheeler, W. C. (2006). The amphibian tree of life. *Bulletin of the American Museum of Natural History, 297*, 1–370. https://doi.org/10.1206/0003-0090(2006)297[0001:TATOL]2.0.CO;2

Comparison:

The laboratory phylogenetic tree also agrees with the classification proposed by Frost et al. (2006), which recognizes major evolutionary lineages among frogs based on molecular evidence. In both the published study and the present analysis, *Xenopus laevis* represents a more distantly related lineage than the other sampled frogs, making it an appropriate outgroup for phylogenetic reconstruction. Furthermore, species within the same genera or closely related genera show greater genetic similarity and tend to cluster together. Differences between the two phylogenies are expected because Frost et al. analyzed multiple genes and a much larger number of species, whereas this laboratory exercise relied only on mitochondrial COI sequences from ten taxa.